

Regression Analysis

Nafiz Imtiaz Rafin

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1 Introduction

This document presents a detailed theoretical overview of **Linear Regression** and **Logistic Regression**, covering mathematical intuition, cost functions, optimization, activation functions, and different types of logistic models. This note is designed to help students and practitioners build a deep understanding of how these models work internally.

2 Linear Regression

Linear Regression is a supervised learning algorithm used for predicting continuous numerical outcomes. It assumes a linear relationship between input features and the output variable.

2.1 Model Representation

Given an input vector:

$$\mathbf{x} = [x_1, x_2, \dots, x_n]$$

Linear regression predicts output $\hat{y}$ using:

$$\hat{y} = w_0 + w_1x_1 + w_2x_2 + \dots + w_nx_n$$

or in vector form:

$$\hat{y} = \mathbf{w}^\top \mathbf{x}$$

Here:

- w_0 is the bias term.
- $w_1, \dots, w_n$ are model weights.

2.2 Assumptions of Linear Regression

Linear Regression works under the following assumptions:

1. **Linearity**: The relationship between features and target is linear.
2. **Independence**: Observations are independent.
3. **Homoscedasticity**: Constant variance of errors.
4. **Normality**: Errors follow a normal distribution.

2.3 Loss Function: Mean Squared Error (MSE)

To learn the parameters, we minimize:

$$J(\mathbf{w}) = \frac{1}{2m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)})^2$$

The factor $\frac{1}{2}$ simplifies gradients.

2.4 Gradient Descent Optimization

We calculate partial derivatives:

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

Then update parameters:

$$w_j := w_j - \alpha \frac{\partial J}{\partial w_j}$$

where α is the learning rate.

2.5 Closed-Form Solution (Normal Equation)

A direct non-iterative solution:

$$\mathbf{w} = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y}$$

This is efficient for small feature sets but computationally expensive for large datasets.

3 Logistic Regression

Logistic Regression is used for **classification problems**. Despite the name, it is a classification algorithm, not regression.

The goal is to estimate the probability:

$$P(y = 1 \mid \mathbf{x})$$

3.1 Why Linear Regression Cannot Be Used for Classification

A regression line can output values outside the range $[0, 1]$, which is invalid for probabilities. Logistic regression fixes this using a **sigmoid activation function**.

3.2 Sigmoid Activation Function

The sigmoid function squashes any real number into $[0, 1]$:

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

where

$$z = \mathbf{w}^\top \mathbf{x}$$

3.2.1 Properties of the Sigmoid Function

- Output range: $(0, 1)$
- Smooth and differentiable
- Useful for binary classification

3.3 Model Representation

Logistic regression outputs:

$$\hat{y} = P(y = 1 \mid \mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x})$$

3.4 Decision Boundary

Prediction:

$$\hat{y} = \begin{cases} 1 & \text{if } \sigma(\mathbf{w}^\top \mathbf{x}) \geq 0.5 \\ 0 & \text{otherwise} \end{cases}$$

3.5 Loss Function: Binary Cross Entropy

A naïve MSE loss fails because it makes the sigmoid non-convex. Therefore, logistic regression uses:

$$J(\mathbf{w}) = -\frac{1}{m} \sum_{i=1}^m [y^{(i)} \log(\hat{y}^{(i)}) + (1 - y^{(i)}) \log(1 - \hat{y}^{(i)})]$$

This loss is convex, ensuring a unique global minimum.

3.6 Gradient of the Loss Function

Derivative:

$$\frac{\partial J}{\partial w_j} = \frac{1}{m} \sum_{i=1}^m (\hat{y}^{(i)} - y^{(i)}) x_j^{(i)}$$

Update rule:

$$w_j := w_j - \alpha \frac{\partial J}{\partial w_j}$$

3.7 Interpretation of Weights

$$\log \left(\frac{P(y=1)}{P(y=0)} \right) = \mathbf{w}^\top \mathbf{x}$$

Weights represent the change in **log-odds** per unit change in feature.

4 Activation Functions in Logistic Models

Logistic Regression itself uses the sigmoid function, but classification models can use other activation functions:

4.1 Sigmoid Function

$$\sigma(z) = \frac{1}{1 + e^{-z}}$$

4.2 Softmax Function (Multiclass)

$$\text{softmax}(z_i) = \frac{e^{z_i}}{\sum_j e^{z_j}}$$

Used in multinomial logistic regression.

4.3 Tanh Function

$$\tanh(z) = \frac{e^z - e^{-z}}{e^z + e^{-z}}$$

Maps values to $[-1, 1]$.

5 Types of Logistic Regression

5.1 1. Binary Logistic Regression

Two-class problems:

$$y \in \{0, 1\}$$

5.2 2. Multinomial Logistic Regression

For K classes:

$$P(y = k \mid x) = \frac{e^{\mathbf{w}_k^\top \mathbf{x}}}{\sum_{j=1}^K e^{\mathbf{w}_j^\top \mathbf{x}}}$$

5.3 3. Ordinal Logistic Regression

Used when categories have an order (e.g., low, medium, high).

Uses **cumulative logits**.

6 Summary

- Linear Regression: Predicts continuous values, optimizes MSE.
- Logistic Regression: Predicts class probabilities, uses sigmoid and cross-entropy loss.
- Logistic regression models the **log-odds** linearly.
- Gradient descent is used to optimize both models.
- Softmax extends logistic regression to multiclass problems.